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RESEARCH

- **Galaxy Formation Model:** *Toward a simulation-motivated and observation-calibrated semi-analytical model.*
- **Galaxy Evolution:** *What drives galaxy quenching, metal enrichment, and morphology transformation?*
- **Galaxy-Halo Connection:** *To what extent are the properties of galaxies shaped by their dark matter halos?*
- **Dark Matter Halo:** *How do halo structure, assembly history, and spatial distribution interplay with each other?*

EXPERIENCE

- Postdoctoral Research Associate, ICC and CEA, **Durham University** since 07/2024
- KIAA Fellow, Kavli Institute for Astronomy and Astrophysics, **Peking University** 07/2022 - 06/2024

EDUCATION

- Ph.D. in Astronomy, **Tsinghua University** 09/2017 - 07/2022
- Visiting Scholar, **University of Massachusetts, Amherst** 11/2019 - 10/2021
- B.S. in Astronomy, **University of Science and Technology of China (USTC)** 09/2013 - 07/2017

PUBLICATION SUMMARY

- ◆ **45** publications (**32** since 2023); **19** as the first/corresponding author (**15** since 2023)
- ◆ **>520** citations (**>425** since 2023); *H*-index: **>15**; open in [NASA/ADS](https://ui.adsabs.org/)

SELECTED PUBLICATION

1. Inflow-driven galaxy evolution - I. Revealing the physics of the fundamental metallicity relation

Kai Wang[†], Carlton Baugh, N. F. Boardman, Sownak Bose, Zheng Cai, Shaun Cole, Carlos S. Frenk, Cedric Lacey, Peder Norberg, Yingjie Peng, Isabel Santos-Santos, Amélie Saintonge, Tom Theuns, Enci Wang
submitted to *MNRAS* (2026) [arXiv: 2608.04784]

I derived a unified framework for the chemical evolution of galaxies from the two mass-continuity equations, providing the first self-consistent explanation of the fundamental metallicity relation (FMR) and its redshift invariance to $z \sim 3$. The FMR is a projection of the gaseous FMR, and its parameterisation encodes how star formation efficiency varies with stellar mass and redshift, and its shape shows the galaxies transit from inflow-driven to equilibrium.

2. Forged in Quenching: Morphological Transformation across Star-forming and Quiescent Galaxies in EAGLE

Kai Wang[†], Carlton Baugh, Sownak Bose, Shaun Cole, Carlos S. Frenk, Hao Fu, Cedric Lacey, Shengdong Lu, Aaron Ludlow, Peder Norberg, Yingjie Peng, Katy L. Proctor, Isabel Santos-Santos, Francesco Shankar, Tom Theuns, Enci Wang, Tao Wang, Vivienne Wild
submitted to *MNRAS* (2026) [arXiv: 2609.16187]

Using EAGLE, I showed that morphology actively drives SMBH growth and quenching central galaxies, rather than being passively inherited through progenitor bias. I also showed that merger-driven SMBH growth in semi-analytic models cannot reproduce the size difference between star-forming and quiescent galaxies, pointing to the need for a secular, morphology-dependent SMBH growth channel.

3. The origin of the galaxy size-stellar metallicity relation: A semi-analytical perspective

Kai Wang[†]

MNRAS 545, 1 (2026) [arXiv: 2510.02573]

Using the semi-analytic framework, I showed that the galaxy size–metallicity anti-correlation persists when the mass-loading factor is controlled, ruling out deeper potentials as the driver, and revealed instead that size regulates star formation efficiency and hence the metallicity.

4. An efficient and robust method to estimate halo concentration based on the method of moments

Kai Wang[†], Houjun Mo, Yangyao Chen, Joop Schaye

MNRAS 527, 10760 (2023) [arXiv: 2310.00200]

I proposed a robust moment-based estimator of halo concentration, now widely adopted in the simulation processing tools, like SOAP and AHE, and in the cosmological simulations, like FLAMINGO and COLIBRE.

5. Dissect two-halo galactic conformity effect: The dependence of star formation activities on the large-scale environment for central galaxies

Kai Wang[†], Yingjie Peng[†], Yangyao Chen

MNRAS 523, 1268 (2023) [arXiv: 2304.06886]

I found that backsplash galaxies explain the dependence of central galaxy quenching on large-scale environment and give rise to two-halo galactic conformity in both IllustrisTNG and SDSS.

GRANTS

- | | |
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| • KIAA fellow start-up research funding
(50,000CNY) | 07/2022 - 07/2024 |
| • China Scholarship for the Visiting Scholar
China Scholarship Council (\$45,600) | 11/2019 - 10/2021 |
| • National Astronomy Training Base
Measure the conditional luminosity functions of galaxies at $z \sim 0.6$ using CLAUDS and BOSS (20,000CNY) | 06/2016 - 06/2017 |
| • National Astronomy Training Base
Thermal gravitational-wave background in the general pre-inflationary scenario (20,000CNY) | 05/2015 - 05/2016 |

MENTORSHIP/SUPERVISION

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|--|-------------------|
| • Oscar Hickman , Postgraduate student at Durham University
Less is More: Unbiased Galaxy Clustering from N-Body Merger Trees | since 02/2026 |
| • Sienna Folkes-Miller , L4 student at Durham University
Tracing the Invisible: How Galaxies Reveal the Dark Universe | since 09/2026 |
| • Thomas Power , L4 student at Durham University
The Dance of Milky Way and Andromeda: A Celestial Ballet Across Cosmologies | since 10/2024 |
| • Thomas Richardson , L4 student at Durham University
Where Shadows Follow Light: Baryon-Induced Dark Matter Halo Contraction in Hydrodynamical Simulations | since 10/2024 |
| • Zeyu Gao , graduate at Peking University
Decoding the SEDs of galaxies with a prior from hydrodynamical simulations [arXiv: 2408.07749] | 11/2022 - 07/2024 |
| • Chengyu Ma , graduate at USTC
Revisiting the fundamental metallicity relation with observation and simulation [arXiv: 2407.21716] | 12/2023 - 07/2024 |
| • Haochen Jiang , undergraduate at USTC
Dissecting the quenching of massive central galaxies in TNG [arXiv: 2510.24886] | 12/2023 - 10/2025 |
| • Xunda Sun , graduate at the University of Chinese Academy of Sciences
Characterizing the spatial distribution of the metal content for galaxies in FIRE2 [arXiv: 2409.09290] | 06/2023 - 10/2024 |
| • Zhijun Zhang , undergraduate at Peking University
Identify protoclusters from high-redshift photometric surveys [Bachelor Thesis] | 09/2022 - 06/2023 |

TEACHING

- **Level-1 Physics Tutorial** Tutor, Durham University, 2024-2025
Year-long weekly tutorials for five undergraduates (mechanics, E&M, quantum mechanics etc.), taught interactively with board work by students. Peer-observed by the Director of Education, Prof. Michael Hunt: "a well planned and conducted tutorial", noting the practice of "going beyond the model answer to explain underlying concepts".
- **ExGal Summer School on Astrophysical Simulations** Tutorial assistant, Durham University, 2025
- **Cosmology and Galaxy Evolution** Guest Lecturer, Peking University, Autumn 2023
- **Observational Cosmology** Teaching Assistant, Tsinghua University, Autumn 2017
- **Particle Cosmology** Teaching Assistant, USTC, Spring 2017
- **General Relativity** Teaching Assistant, USTC, Autumn 2016

OUTREACH

- **Celebrate Science, The Universe Makers Team** Royal Society, London, 2026
The Summer Science Exhibition is the Royal Society's flagship annual public engagement festival celebrating the cutting edge of UK research and innovation. Welcoming over 10,000 total visitors—including 2,000 secondary school students and teachers—the high-capacity, six-day event features 13 interactive flagship exhibits and over 60 specialised talks and panels delivered by 100+ world-leading speakers.
- **Undergraduate Open Day, Demonstrator** Durham University, 2026
The Undergraduate Open Day is Durham University's flagship two-day recruitment event for prospective applicants and parents, showcasing its top-5 UK standing. Spanning 16 undergraduate colleges and dozens of departments, the event utilises a massive staff and student ambassador network to deliver presentations, exhibitions, and campus tours.
- **Schools' Science Festival, Demonstrator** Durham University, 2025
The Durham University Schools' Science Festival is an annual outreach program designed to inspire Year 9 and Year 10 students Durham University Schools' Science Festival 2025. The 2025 festival welcomed over 700 pupils from 22 schools across the region to participate in interactive, hands-on scientific projects.
- **Celebrate Science, Demonstrator** Palace Green, Durham, 2024
Celebrate Science is Durham University's annual public festival aimed at children aged 7–11, families, and school groups, designed to make complex research accessible. The event attracts over 5,000 to 7,500+ attendees annually, featuring over 200 volunteers to deliver interactive STEM workshops.

SERVICE

- **Referee** for MNRAS (since 2024), ApJ (since 2023), and A&A (since 2022)
- **SOC member** and **discussion panelist** of session in NAM 2026 2026
The NAM 2026 session Galaxy Chemistries Across Cosmic Time featured 2 invited talks, 10 contributed talks, 7 poster presentations, and a panel discussion, bringing together observers and theorists to discuss galaxy chemical evolution from the local Universe to the highest redshifts.
- **SOC member** of session "The Golden Era of Gravitational Lensing" of NAM 2025 2025
The NAM 2025 session The Golden Era of Gravitational Lensing featured 18 contributed talks and 4 poster presentations, covering strong and weak lensing, microlensing, gravitational-wave lensing, dark matter, cosmology, and next-generation surveys including Euclid, JWST, Rubin LSST, Roman, and SKA.
- **Organiser** of CosmICConnections at Durham University 2026
CosmICConnections is a hybrid seminar and discussion series held every four weeks, providing an informal forum for local and visiting researchers to present their work, exchange ideas, and foster collaborations across astronomy and cosmology.
- **Organiser** of Friday Lunchtime Astronomy Talks (FLAT) at Durham University 2025-2026
The Friday Lunchtime Astronomy Talk (FLAT) is a weekly seminar series featuring around 60 talks annually by local and visiting early-career researchers, fostering scientific discussion and collaboration among students and postdoctoral researchers.

- **LOC member** of National Astronomy Meeting (NAM) 2025 at Durham University 2025
NAM 2025, hosted by Durham University, brought together over 1,400 attendees across scientific and public events, featured more than 500 talks, and strengthened collaboration across the UK astronomy community while showcasing Durham as an international hub for astronomy research.
- **Co-lead** of morning arXiv journal club at Durham University 2025
- **Member** of Faculty Candidate Interview Committee at KIAA, Postdoc Representative 2023, 2024
- **Organiser** of weekly Galaxy Party at KIAA 2023
- **Organiser** of the Postdoc Science Day at KIAA 2022
- **Organiser** of the Speaker Lunch at the Tsinghua Center for Astrophysics 2018-2019

SEMINAR TALKS

- DEEP-Theory meeting, University of California, Santa Cruz
From inflow-driven To equilibrium: A unified picture of the fundamental metallicity relation Remote, Aug. 2026
- Astro Seminar at Kavli-IPMU, University of Tokyo (**invited**)
From inflow-driven To equilibrium: How the baryon cycle shapes the metallicity of galaxies Remote, Aug. 2026
- Seminar of Galaxy and Cosmology at National Astronomical Observatories, CAS (**invited**)
Forged in quenching: An EAGLE's view of morphological transformation Beijing, Apr. 2026
- KIAA-DoA Seminar at Peking University (**invited**)
Forged in quenching: An EAGLE's view of morphological transformation Beijing, Apr. 2026
- Informal Talk at Tsinghua University (**invited**)
Metallicity as an Astrophysical Diagnostic Beijing, Apr. 2026
- Astronomy Seminar, Nanjing University (**invited**)
Forged in quenching: An EAGLE's view of morphological transformation Nanjing, Apr. 2026
- Astronomy Seminar, Westlake University (**invited**)
How well do we understand galaxy formation before JWST challenges it? Hangzhou, Apr. 2026
- Astrophysics Research Institute Seminar, Liverpool John Moores University (**invited**)
Metallicity as an Astrophysical Diagnostic Liverpool, Nov. 2025
- Lunchtime Talks at Astronomy Group, University of St Andrews (**invited**)
Metallicity as an Astrophysical Diagnostic St Andrews, Oct. 2025
- Informal Talk at Tsinghua University (**invited**)
Local Group Analogs in a cosmological context Beijing, Jun. 2025
- KIAA-DoA Seminar at Peking University (**invited**)
Towards the next-generation semi-analytical galaxy formation model Beijing, Jun. 2025
- USTC Astronomy Seminar Series (**invited**)
Towards the next-generation semi-analytical galaxy formation model Hefei, Jun. 2025
- DoA Seminar at Shanghai Jiao Tong University (**invited**)
Towards the next-generation semi-analytical galaxy formation model Shanghai, May 2025
- Friday Lunchtime Astronomy Talks at Durham University
Local Group Analogs in a cosmological context Durham, Mar. 2025
- Galaxy & Cosmology seminar at Tsinghua University (**invited**)
Dark matter halo and its structure, assembly, and clustering Beijing, May 2024
- Lunch talk at South-Western Institute For Astronomy Research, Yunnan University (**invited**)
Galaxy formation within and without dark matter halos Kunming, Apr. 2024
- Astronomy Seminar at the Huazhong University of Science and Technology (**invited**)
Galaxy formation within and without dark matter halos Wuhan, Mar. 2024
- ITC Luncheon at the Center for Astrophysics | Harvard & Smithsonian (**invited**)
How to connect galaxies across cosmic time? Cambridge, Jan. 2024
- Steward/NOIRLab Galaxy Group Talk
How to connect galaxies across cosmic time? Tucson, Jan. 2024
- Carnegie arXiv Tea (**invited**)
Relating galaxies across different redshift to study galaxy evolution Pasadena, Jan. 2024

- KIPAC tea talk at Stanford University (**invited**)
Characterizing the assembly of dark matter halos with protohalo size histories Stanford, Jan. 2024
- UC Santa Cruz CGI (Cosmology/Galaxies/IGM) Seminar (**invited**)
Central Galaxy Quenching and its relation to Halo Formation & Large-scale Environment Santa Cruz, Jan. 2024
- Galread: Princeton/IAS Galaxy Journal Club
Characterizing the assembly of dark matter halos with protohalo size histories Remote, Oct. 2023
- UC Santa Cruz CGI (Cosmology/Galaxies/IGM) Seminar
Characterizing the assembly of dark matter halos with protohalo size histories Remote, Oct. 2023
- KIAA-DoA Seminar, Peking University (**invited**)
Central Galaxy Quenching and its Relation to Halo Formation Time & Large-scale Environment Beijing, Mar. 2023
- Lunch Talk at the Department of Astronomy, Tsinghua University (**invited**)
Relating galaxies across different redshift Beijing, Nov. 2022
- Lunch Talk at Kavli-IPMU, University of Tokyo (**invited**)
Finding proto-clusters to trace galaxy evolution Remote, Jun. 2021

CONFERENCE TALKS

- DEX-XXII: 22nd Durham-Edinburgh eXtragalactic Workshop
An EAGLE's view of the galaxy size evolution across cosmic time Edinburgh, Jan. 2026
- SWIFTCOON 2025
Dissecting the scatter of stellar mass-halo mass relation in COLIBRE Leiden, Oct. 2025
- L-GALAXIES workshop 2025
GALFORM++: Next-generation semi-analytical galaxy formation model Berlin, Oct. 2025
- National Astronomy Meeting (NAM)
Contributed talks: Durham, Jul. 2025
 - ◆ Local Group Analogs in a cosmological context: Relating the velocity structure to the cosmic web
 - ◆ Testing galaxy formation models with the stellar mass-halo mass relations for star-forming and quiescent galaxies
 - ◆ Environmental Dependence of the Mass–Metallicity Relation in Cosmological Hydrodynamical Simulations**Poster:**
 - ◆ Dissecting two-halo galactic conformity effect for central galaxies
- Expanding the boundaries of dark matter halo
Local Group Analogs in a cosmological context Shanghai, May 2025
- Mock Barcelona 2024 (**invited**)
Stellar mass-halo mass relation to the second order Barcelona, Oct. 2024
- Conference of the Co-evolution of galactic eco-systems and their large-scale environments (**invited**)
Dissecting two-halo galactic conformity effect for central galaxies Hangzhou, Apr. 2024
- The 2nd Shanghai Assembly on Cosmology and Structure Formation
Characterizing the assembly of dark matter halos with protohalo size histories Shanghai, Oct. 2023
- Collaboration Workshop on Cosmology and Galaxy Formation
Relating Galaxies across Cosmic Time to study galaxy evolution Shanghai, Jun. 2023
- 25th Chinese Astronomical Society Guoshoujing Symposium on Galaxies and Cosmology
Central Galaxy Quenching and its Relation to Halo Formation & Large-scale Environment Huangshan, May 2023
- Conference of Star Formation and Nuclei Activity in Galaxies
Central Galaxy Quenching and its Relation to Halo Formation & Large-scale Environment Nanjing, Mar. 2023
- The 11-th Prime Focus Spectrograph collaboration meeting
Identifying galaxy groups from high-z and incomplete spectroscopic surveys Pasadena, Dec. 2019
- The 10-th Prime Focus Spectrograph collaboration meeting
Finding groups/clusters of galaxies in the PFS galaxy evolution survey Shanghai, Dec. 2018